

Healthcare Information Assistant - Design Document

1. System Overview

The Healthcare Information Assistant is a Retrieval-Augmented Generation (RAG) pipeline designed to answer health-related queries using evidence retrieved from trusted medical sources like MedQuad, NICE etc. The system has been designed such that every answer is grounded in retrieved evidence, sources are cited inline, and a confidence score, retrieval accuracy metrics and hallucination score is computed for each response.

2. Knowledge Base

1. Primary Dataset - MedQuAD

MedQuAD (Medical Question Answering Dataset) contains approximately 47,000 question-answer pairs collected from trusted medical organisations including NIH, NCI, NIDDK, and similar bodies. Each entry includes a question, answer, source URL, and focus area (e.g. symptoms, treatment, prevention).

Each Q&A pair is loaded as a LangChain Document with metadata fields: paper (source organisation), number (row index), focus_area (main topic of the question), and type="MedQuAD".

2. Supplementary Guidelines

PDF documents from authoritative medical bodies are loaded from a local guidelines directory. These include WHO Hypertension Guidelines, CDC Heart Health Recommendations, and NICE Hypertension Guidelines. Each PDF page is stored as a Document object with metadata: paper (filename), page (page number), and type="Guideline".

The ingestion pipeline is as follows:

- MedQuAD CSV loaded with pandas, converted to LangChain Documents
- PDFs loaded with PyMuPDFLoader via LangChain DirectoryLoader
- All documents combined and chunked with RecursiveCharacterTextSplitter (chunk_size=600, overlap=70)
- Chunks embedded with BAAI/bge-small-en-v1.5 (normalized embeddings)
- Stored in a FAISS flat index, persisted to disk, so that it doesn't get rebuilt every time the chatbot runs

3. System Components

1. Safety Layer

Before any retrieval occurs, the query is checked against two keyword lists:

- Emergency keywords: suicide, overdose, chest pain, heart attack, stroke, cannot breathe, kill myself, dying, unconscious, want to die. These trigger an immediate emergency services redirect.
- Unsafe keywords: prescribe, diagnose me, what dose, how much medication, can i take, should i take, diagnosis. These trigger a warning and block the query from proceeding.

This layer is conservative so that any suspicious or malicious query is not met with improper advice or hallucinations from the chatbot regarding medical issues that could be harmful.

2. Retrieval and Reranking

Queries are embedded using the same BAAI/bge-small-en-v1.5 model used at ingest time. FAISS `similarity_search_with_score` returns the top 6 most similar chunks using L2 distance along with the score on normalized embeddings (equivalent to cosine similarity). Both MedQuAD and guideline chunks are searched simultaneously from the same FAISS index. The top 6 retrieved chunks are reranked using cross-encoder/ms-marco-MiniLM-L-6-v2, a CrossEncoder model that reads the full query and each chunk together (rather than comparing embeddings separately). This produces more accurate relevance scores than embedding similarity alone. The top 3 chunks after reranking are passed to the confidence score generation step.

3. Confidence Estimation

A confidence score (0–100%) is computed from the CrossEncoder rerank scores using linear normalisation. Cross-encoder scores for this model range approximately from -10 to +10, as cross-encoder logits usually fall in this range. The formula maps this range to a percentage:

$$\text{confidence} = \max(0, \min(100, (\text{avg_score} + 10) / 20 \times 100))$$

Scores above 70% are displayed as high confidence, 40–70% as moderate, and below 40% as low. These thresholds are heuristic.

Softmax to make probability scores was tried but not used in the final version, as in the case of all chunks having high reranked scores, Softmax would flatten them out into similar scores, so the confidence score is not estimated clearly.

4. Answer Generation

The top 3 reranked chunks are assembled into a context block with [Source: paper_name] prefixes. This context, the user query, and a structured prompt are sent to Gemini 2.5 Flash. The prompt instructs the model to:

- Answer only from the provided context
- Cite sources inline (e.g. "According to MedQuAD...")
- Separate facts from assumptions
- Never recommend specific medications, dosages, or diagnoses
- Always recommend consulting a healthcare professional
- Express uncertainty clearly when confidence is low (injected via `confidence_note`)

5. Evaluation Pipeline

Retrieval Evaluation

- Top Chunk Score: Raw CrossEncoder score of the highest-ranked chunk
- Retrieval Success: True if top score > 0
- Precision@K: Fraction of top K chunks with score > 0 (relevant threshold)
- Mean Reciprocal Rank: Shows how high up the first relevant chunk is in the reranking.

CrossEncoder scores above 0 are treated as relevant based on the ms-marco model training convention.

Hallucination Detection

Hallucination risk is estimated using NLI (Natural Language Inference) with cross-encoder/nli-MiniLM2-L6-H768, which is a general-purpose crossencoder trained on a medical dataset. For each of the 3 reranked chunks, the model checks whether the generated answer is entailed by that chunk. The maximum entailment score across all chunks is used as the primary grounding signal.

- Max Entailment > 70%: Low hallucination risk: answer is strongly supported by at least one chunk
- Max Entailment 40-70%: Moderate risk: partial support (neutral)
- Max Entailment < 40%: High risk: answer may not be grounded or rightly present in retrieved evidence (contradiction)

NLI is run per-chunk rather than on the concatenated context to stay within the model's 512-token limit and to avoid truncating evidence.

4. Known Limitations

- Confidence and hallucination thresholds are heuristic, not statistically calibrated against ground truth answer correctness
- NLI entailment tends to underestimate grounding and give moderate or high risk for nondeterministic answers that are enriched with rephrasing and a combination of context over multiple chunks, which is what an LLM usually provides.
- Safety keyword matching is not foolproof; it can be defeated by paraphrasing and may produce false positives on legitimate queries containing different words or worded in a misleading manner.
- No disambiguation for vague queries: "what causes pain" retrieves broadly.

6. Example Queries and Expected Behaviour

"What lifestyle changes help with hypertension?"

Expected: retrieves from medical guidelines like WHO/NICE and MedQuAD hypertension entries, generates a grounded answer citing both sources, high confidence, low hallucination risk.

"I am having chest pain."

Expected: safety layer intercepts, emergency message shown, no retrieval occurs.

"What dose of metformin should I take?"

Expected: unsafe keyword "what dose" detected, warning shown, no retrieval occurs.

"What are the symptoms of type 2 diabetes?"

Expected: MedQuAD entries on diabetes symptoms retrieved, Gemini generates a cited answer, moderate-to-high confidence depending on chunk relevance with moderate to high hallucination risk, as specific information on type 2 diabetes is not present.

7. References

- Ben Abacha, A. et al. (2019). MedQuAD: A Medical Question Answering Dataset. arXiv:1901.08079
- WHO. (2023). Hypertension Guidelines.

- CDC. Heart Disease and Stroke Prevention Resources.
- NICE. (2023). Hypertension in Adults: Diagnosis and Management (NG136).